

Customer Churn Prediction Using Elastic Net Regularized Logistic Regression

Based on the uploaded project: Customer Churn Prediction Using Elastic Net Regularized Logistic Regression

Abstract

Customer churn is a major business problem in subscription-driven and service-oriented industries. This paper presents an interpretable customer churn prediction framework based on Elastic Net regularized logistic regression. The approach combines L1 regularization, which promotes sparsity and feature selection, with L2 regularization, which stabilizes estimates in the presence of correlated predictors. The framework is designed for customer data containing demographic, billing, tenure, subscription and usage attributes. The proposed workflow includes preprocessing, train-test splitting, cross-validation for tuning the regularization strength and mixing parameter, model training, and evaluation using accuracy, precision, recall, F1-score and ROC-AUC. The approach aims to reduce overfitting while retaining a level of interpretability suitable for business decision-making.

Introduction

Customer churn prediction enables organizations to identify customers who are likely to discontinue a service early enough for retention interventions. Traditional rule-based approaches may overlook relationships among multiple customer attributes, while highly complex models can be difficult for business users to interpret. This study therefore focuses on Elastic Net logistic regression as a compromise between predictive modeling and interpretability.

Methodology

The methodology begins by handling missing values, encoding categorical variables and standardizing numerical variables. The dataset is divided into training and testing subsets. Cross-validation is then used to tune λ , the regularization strength, and α , the L1/L2 mixing parameter. The resulting Elastic Net logistic regression model produces churn probabilities and class labels. Performance is assessed using accuracy, precision, recall, F1-score and ROC-AUC, with plain logistic regression, decision trees and random forests serving as benchmark models.

Expected Results and Business Value

The project framework expects Elastic Net to provide stable coefficient estimates while automatically reducing the influence of less useful predictors. Important churn drivers identified in the source project include contract type, monthly charges and customer tenure. The resulting model can provide ranked and interpretable indicators of customer risk and can be incorporated into a CRM process for proactive retention.

Conclusion

Elastic Net regularized logistic regression provides a practical framework for churn prediction when customer features are numerous and correlated. Its combination of feature selection and coefficient stabilization supports both generalization and interpretability. Future empirical work should report the tuned parameters and measured test-set performance on the selected dataset.

References / Source Basis

1. Uploaded project presentation, "Customer Churn Prediction Using Elastic Net Regularized Logistic Regression," A.Y. 2026–27. The presentation describes the business problem, Elastic Net rationale, methodology pipeline, evaluation metrics, Telco Customer Churn dataset and related literature.
2. Arivazhagan, B. & Sankara Subramanian, R.S. "Customer Churn Prediction using Logistic Regression with Regularization and Optimization Technique." IJITEE, Vol. 9, Issue 9, 2020.
3. Šimovič, P. P., Chen, C. Y. T. & Sun, E. W. "Classifying the Variety of Customers' Online Engagement for Churn Prediction with a Mixed-Penalty Logistic Regression." Computational Economics, 2023.
4. Yang, L., Li, D. & Lu, Y. "Prediction Modeling and Analysis for Telecom Customer Churn in Two Months." arXiv:1911.00558, 2019.